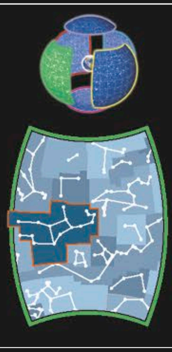
THE VIRGIN

Virgo



SIZE RANKING 2

BRIGHTEST STAR
Spica (α) 1.0

GENITIVE Virginis

ABBREVIATION Vir

HIGHEST IN SKY AT 10 P.M.
April–June

FULLY VISIBLE
67°N–75°S

Virgo straddles the celestial equator, between Leo and Libra. It is the largest constellation of the zodiac, and the second-largest overall. The constellation depicts a Greek virgin goddess (see panel, right). Virgo contains the Virgo Cluster (see p.329), the closest large cluster of galaxies to Earth, which is some 50 million light-years away and which extends over the border of Virgo into Coma Berenices. The Sun is in Virgo during the September equinox each year.

SPECIFIC FEATURES

Gamma (γ) Virginis, or Porrima (see p.253), is a binary star with the relatively short period of 169 years. As a result of this short period, the effects of the two stars' orbital motions can easily be followed through amateur telescopes. As seen from Earth, the two stars were closest together in 2005, when a telescope with an aperture of 10 in (250 mm) was needed to separate them. By 2012, the stars had moved far enough apart that they could be divided by a telescope of only 2.4 in (60 mm). For the rest of the 21st century, it will be possible to split the components of Gamma Virginis with a small-aperture telescope. Both of the stars are of magnitude 3.5.

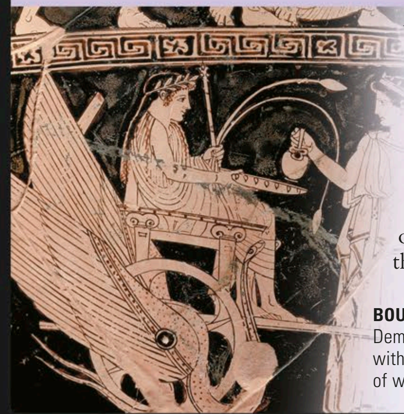
In the upper part of Virgo's body lie the numerous galaxies of the Virgo Cluster. None is easy to see with a small instrument. The brightest members are giant ellipticals, notably M49, M60 (see p.317), M84, M86, and M87 (see p.323). M87 is a strong radio and X-ray source also known as Virgo A. Long-exposure photographs show it is ejecting a jet of gas, like certain quasars.

The Sombrero Galaxy (see p.316), or M104, is Virgo's best-known galaxy. This spiral is about two-thirds as far away as the Virgo Cluster. It is oriented almost edge-on to the Earth, so that a dark lane of dust in the galaxy's plane crosses its central bulge. The bulge may be all that can be seen through a small telescope; the dust lane is only revealed when seen through a large-aperture telescope or on long-exposure photographs.

The brightest quasar in the sky, 3C 273 (see p.325), also lies in the bowl of Virgo. However, it is much more distant than the Virgo Cluster. Through most telescopes, it appears as nothing more than a 13th-magnitude star. Only professional equipment will reveal it as the center of an active galaxy, which is some 2,000 million light-years away from Earth.

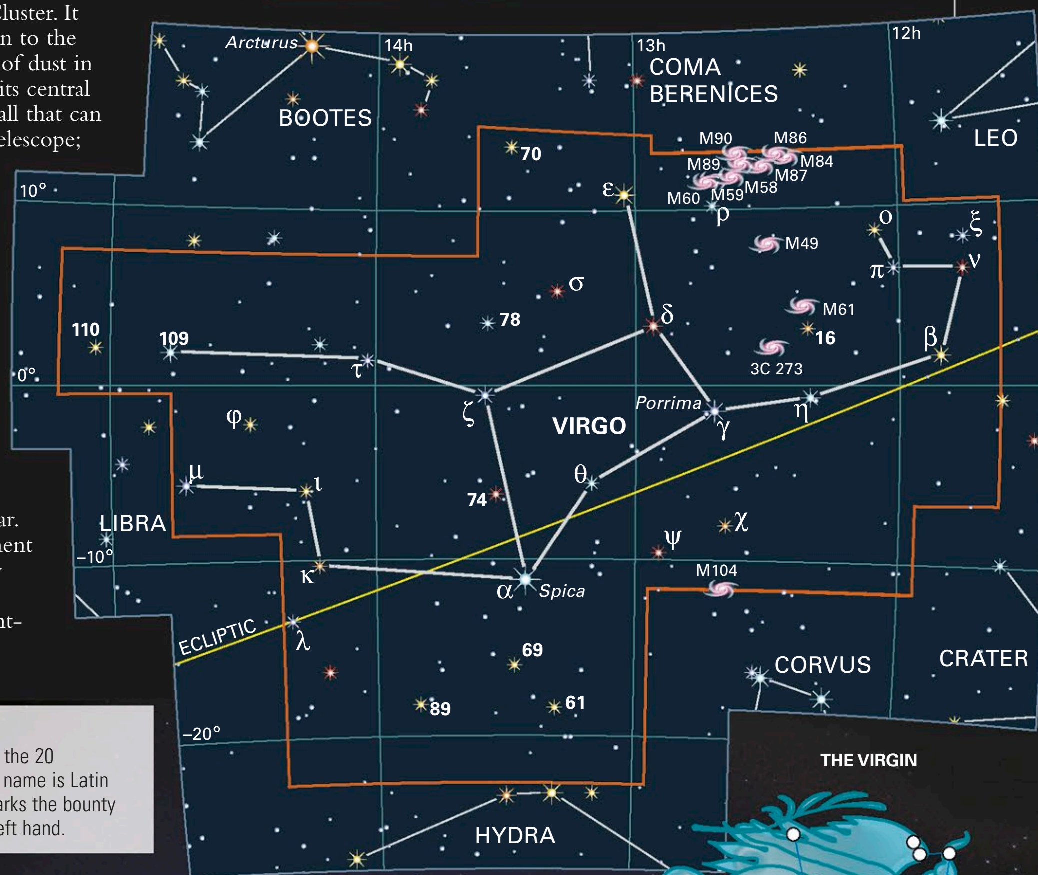
MYTHS AND STORIES

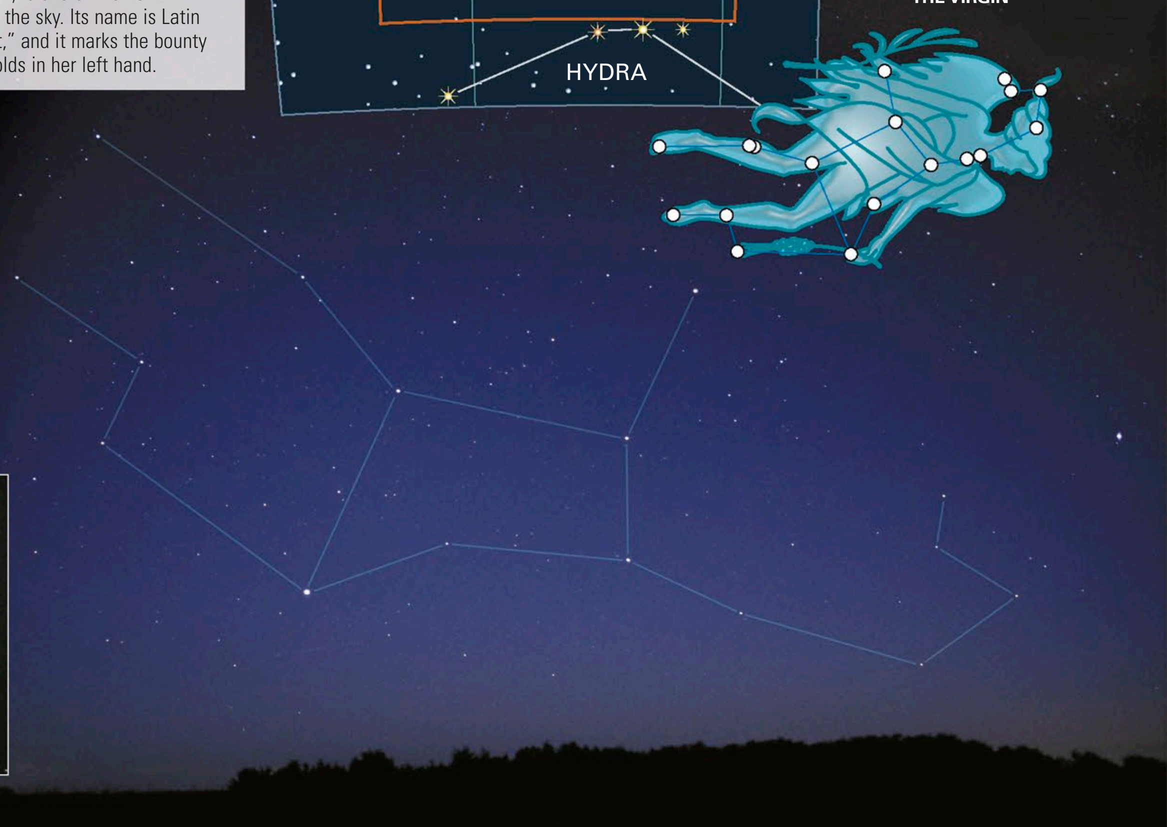
THE VIRGIN GODDESS



Virgo is usually identified as Dike, the Greek goddess of justice, who abandoned the Earth and flew up to heaven when human behavior deteriorated. Neighboring Libra represents her scales of justice. Virgo is also visualized as Demeter, the harvest goddess, who holds an ear of wheat, which is represented by the constellation's brightest star, Spica.

BOUNTIFUL OFFERINGS
Demeter presented Triptolemus, a prince of Eleusis, with a chariot drawn by winged dragons and grains of wheat to sow crops wherever he traveled.





THE WHEAT GODDESS
Spica (bottom, left) is one of the 20 brightest stars in the sky. Its name is Latin for "ear of wheat," and it marks the bounty that the Virgin holds in her left hand.

THE SOMBRERO GALAXY
The Sombrero Galaxy (M104) is a spiral galaxy with a large central bulge, seen almost edge-on, and resembling a Mexican hat. It lies about 30 million light-years away.

M87
Through a small telescope, the giant elliptical galaxy M87 appears as a rounded glow, but photographs and CCD images reveal the jet of gas that is being expelled from its highly active nucleus. Here, the jet is just visible near the top right of the core.